Summary in Graph

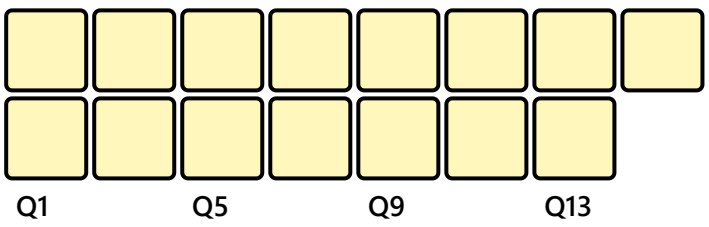
Exam Summary (GO Classes GATE CS/DA | DBMS | Normal Forms | Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (61) out of 605 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Databases

Match the following :

(1)	Determinants	(a)	No attribute can be added
(2)	Candidate key	(b)	Uniquely identified a row
(3)	Non-redundancy	(c)	A constraint between two attribute
(4)	Functional Dependency	(d)	Group of attributes on the left hand side of arrow of function dependency.

A. 1-d, 2-b, 3-a, 4-c
B. 2-d, 3-a, 1-b, 4-c
C. 4-a, 3-b, 2-c, 1-d
D. 3-a, 4-b, 1-c, 2-d

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Which of the following statements is true?

D_1: The decomposition of the schema R(A, B, C) into R_1(A,B) and R_2(A,C) is always lossless

D_2: The decomposition of the schema R(A, B, C, D, E) having AD $\rightarrow$ B, C $\rightarrow$ DE, B $\rightarrow$ AE, and AE $\rightarrow$ C, into R_1 (A,B, D) and R_2(A, C, D, E) is lossless

A. Both D_1 and D_2
B. Neither D_1 nor D_2
C. Only D_1
D. Only D_2

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Which one is correct w.r.t. RDBMS?

A. primary key $\subseteq$ super key $\subseteq$ candidate key
B. primary key $\subseteq$ candidate key $\subseteq$ super key
C. super key $\subseteq$ candidate key $\subseteq$ primary key
D. super key $\subseteq$ primary key $\subseteq$ candidate key

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

In relational database management, which of the following is/are property/properties of candidate key?

P: Uniqueness

Q: Irreducibility

A. P only
B. Q only
C. Both P and Q
D. Neither P nor Q

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Suppose R is a relation schema and F is a set of functional dependencies on R. Further, suppose R1 and R2 form a decomposition of R. Then the decomposition is a lossless join decomposition of R provided that:

A. $R1 \cap R2 \rightarrow R1$ is in F+
B. $R1 \cap R2 \rightarrow R2$ is in F+
C. both $R1 \cap R2 \rightarrow R1$ and $R1 \cap R2 \rightarrow R2$ functional dependencies are in F+
D. at least one from $R1 \cap R2 \rightarrow R1$ and $R1 \cap R2 \rightarrow R2$ is in F+

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Which of the following is true?

A. A relation in BCNF is always in 3NF
B. A relation in 3NF is always in BCNF
C. BCNF and 3NF are same
D. A relation in BCNF is not in 3NF

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The third normal form is based on the concept of____

A. Closure Dependency
B. Transitive Dependency
C. Normal Dependency
D. Functional Dependency

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Armstrong (1974) proposed systematic approach to derive functional dependencies. Match the following w.r.t functional dependencies:

	List-I		**List -II**
a.	Decomposition Rule	i.	If $X \rightarrow Y$ and $Z \rightarrow W$ then $\{X, Z\} \rightarrow \{Y, W\}$
b.	Union rule	ii.	If $X \rightarrow Y$ and $\{Y, W\} \rightarrow Z$ then $\{X, W\} \rightarrow Z$
c.	Composition rule	iii.	If $X \rightarrow Y$ and $X \rightarrow Z$ then $X \rightarrow \{Y, Z\}$
d.	Psedudo transitivity rule	iv.	If $X \rightarrow \{Y, Z\}$ then $X \rightarrow Y$ and $X \rightarrow Z$

Codes:

A. a-iii, b-ii, c-iv, d-i
B. a-i, b-iii, c-iv, d-ii
C. a-ii, b-i, c-iii, d-iv

D. a-iv, b-iii, c-i, d-ii

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider a relational schema $S = (U, V, W, X, Y, Z)$ on which the following functional dependencies hold:

$(U \rightarrow V, VW \rightarrow X, Y \rightarrow W, X \rightarrow U)$

Which are the candidate keys among following options?

A. UY, VY
B. UY, VY, XY
C. UYZ, VYZ, VWZ
D. UYZ, VYZ, XYZ

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

The best normal form of relation scheme $R(A, B, C, D)$ along with the set of functional dependencies $F = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B\}$ is

A. Boyce-Codd Normal form
B. Third Normal form
C. Second Normal form
D. First Normal form

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Identify the minimal key for relational scheme $R(A, B, C, D, E)$ with functional dependencies $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$

A. A
B. AE
C. BE
D. CE

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Let $R = ABCDE$ is a relational scheme with functional dependency set $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$. The attribute closures of A and E are

A. $ABCD, \phi$
B. $ABCD, E$
C. Φ, ϕ

D. ABC, E

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

A relation R={A,B,C,D,E,F,G} is given with following set of functional dependencies: F={AD→E, BE→F, B→C, AF→G}. Which of the following is a candidate key?

A. A
B. AB
C. ABC
D. ABD

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

For a database relation $R(a, b, c, d)$, where the domains a, b, c, d include only atomic values, only the following functional dependencies and those that can be inferred from them hold

- $a \rightarrow c$
- $b \rightarrow d$

This relation is

A. in first normal form but not in second normal form

B. in second normal form but not in first normal form

C. in third normal form

D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following relational schemes for a library database:

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Book (Title, Author, Catalog_no, Publisher, Year, Price)
Collection(Title, Author, Catalog_no)
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with the following functional dependencies:

I. $\text{Title Author} \rightarrow \text{Catalog_no}$

II. $\text{Catalog_no} \rightarrow \text{Title Author Publisher Year}$

III. $\text{Publisher Title Year} \rightarrow \text{Price}$

Assume $\{\text{Author, Title}\}$ is the key for both schemes. Which of the following statements is true?

A. Both Book and Collection are in BCNF

B. Both Book and Collection are in 3NF only

C. Book is in 2NF and Collection in 3NF

D. Both Book and Collection are in $2\mathrm{NF}$ only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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